The quadratic formula

1

a $a = 1, b = 7, c = 10$

b $a = 1, b = -3, c = -4$

c $a = 2, b = 9, c = 0$

d $a = 4, b = 3, c = -5$

e $a = 8, b = -3, c = 0$

f $a = -8, b = 3, c = 0$

g $a = 2, b = -9, c = -5$

h $a = 3, b = -17, c = 9$

2

a $x = -2, -3$

b $x = 2, 3$

c $x = \frac{9}{2}, \frac{1}{2}$

d $x = \frac{11}{2}, \frac{5}{2}$

e $x = -4, 1$

f $x = \frac{1}{2}, 2$

g $x = -\frac{3}{4}, -1$

h $x = 5, -\frac{3}{4}$

i $x = \frac{3}{5}, -2$

j $x = 3, -\frac{2}{5}$

k $x = 5, -\frac{4}{3}$

l $x = \frac{4}{5}, -5$

3

a $x = \frac{7 \pm \sqrt{13}}{2}$

b $x = \frac{5 \pm \sqrt{33}}{2}$

c $x = \frac{-15 \pm \sqrt{193}}{4}$

d $x = \frac{-9 \pm \sqrt{129}}{6}$

e $x = \frac{15 \pm \sqrt{185}}{10}$

f $x = \frac{-15 \pm \sqrt{285}}{10}$

g $x = \frac{17 \pm \sqrt{469}}{6}$

h $x = \frac{2 \pm \sqrt{29}}{15}$

4

a $x = \pm\sqrt{7}$

b $x = \pm 2.646$

5

a $x = 1.372, -4.372$

b $x = 0.405, -7.405$

c $x = -0.360, -1.390$

d $x = -0.260, -1.540$

e $x = 5.589, -0.089$

f $x = 0.254, -5.254$

g $x = 0.219, -0.965$

h $x = 0.390, -3.279$

6 $m = 7 \pm 4\sqrt{3}$

7

a

i $a = -7, b = 5, c = 10$

ii $-7x^2 + 5x + 10 = 0$

b

i $a = 9, b = 22, c = 3$

ii $9x^2 + 22x + 3 = 0$

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Applications

8 $x = 7.1$

9 $6\ \text{ms}^{-1}$

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